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Building proactive law competence with systems intelligence

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Abstract: Proactive law focuses on preventing future problems and promoting desirable outcomes through the application of law. This article argues that framing proactive law as a systemic approach opens up the possibility for developing a proactive legal theory and solutions using insights from various applications of systems thinking, which is an aspect currently missing from proactive law research. The article introduces the framework of systems intelligence (SI), a practically oriented philosophy for making positive systemic interventions, and discusses its promises and challenges in building proactive law competence. The SI framework is used to illustrate how formalistic, legal-centric and problem-focused practices of law tend to create systemic "reactive-law generators" that may contribute to reproducing the problems they were initially designed to solve. The paper argues that understanding the mechanisms of the reactive-law generators and mitigating their negative effects should be the focus of proactive law research and suggests that systems intelligence philosophy, as well as other systems-awareness building methods, such as visual and co-creative design techniques, can provide strategies and tools to support this work.

Keywords: *legal design, proactive law, sustainability, systems awareness, systems intelligence, systems theory, transformation*

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1. Introduction

We live in a world of systems. Their subtle patterns of connection are all around us: in the way our eyes and brain work together to construct a perception of this text, in the way writing this article builds on centuries of accumulated human knowledge, and in the way you and I connect with each other through the practice of academic reading and writing. According to systems theories, everything in the world, whether human-made or of natural origin, is part of an interconnected whole (Capra and Luisi, 2014, p. 80). At some level we can all sense this. Whenever we notice repeating behavioral patterns between two seemingly separate entities, we often intuitively understand that there is a system at play (Edson, Tuddenham and Buckle-Henning, 2016, p. 16). Mobus and Kalton (2015, p. 131) note that this is due to our brains' evolutionary adaptation to the systemic operations of nature. In other words, we are all born as systems thinkers.

Most of the time, however, the complexity of our systemic environment exceeds what we can possibly sense and understand, making life unpredictable and beyond our control. Although systems are everywhere, often right in front of our eyes, we tend to ignore them, see only their surface, or attempt to interpret their complex interconnections and feedback mechanisms through siloed knowledge. This tendency is further amplified by Western education, which, over centuries, has fostered an atomistic, Newtonian view of the world. Since the emergence of modern science in the seventeenth century, scientific disciplines, including law, have specialized in understanding the parts of a whole rather than their interrelations (Mobus and Kalton, 2015, p. 10). Systems blindness, combined with atomistic thinking and our inherited human bias to assume that our personal views align with those of others, explains why our endeavors sometimes fail, despite our good intentions and best efforts (Midgley, 2000, p. 11; Pronin, 2008, p. 1177).

The twenty-first–century polycrisis is the ultimate consequence of humanity's collective systems blindness. A myriad of pressing environmental, social and economic problems, to name a few, have emerged from intertwined feedback loops where short-term gains are favored over long-term sustainability. The polycrisis also represents the ultimate challenge, demanding that we leverage and expand our natural systems thinking abilities. According to the philosophy of *systems intelligence*, our innate ability to sense systems enables us to pursue possibilities even when the external reality appears to

conflict with this potential (Hämäläinen, Jones and Saarinen, 2014, p. 35). Raimo Hämäläinen and Esa Saarinen (2006, p. 191), the developers of the concept of systems intelligence, suggest that becoming intentionally more aware of the systems around us, including the social systems we personally take part in, may empower us to act more positively and purposely within them. This view is generally shared also in proactive law (Pohjonen, 2006, p. 55).

Proactive law can be characterized as a legal mindset and a future-oriented practical approach promoting desirable economic and social goals through law. It emphasizes the potential of law to work as an enabler of positive outcomes, creating trust, legal certainty and sustainability to social relationships, such as between businesses, their customers and other stakeholders (European Economic and Social Committee, 2008). To achieve these goals, proactive law aims to produce knowledge on how legal problems can be prevented *ex ante* before they occur, and, in particular, how desirable outcomes can be promoted and achieved through legal practices (Berger-Walliser, 2012, p. 14; Haapio, 2006, p. 23). A key argument of the approach is that to prevent problems from recurring, law must have tools to tackle the root causes of the problems, which are often "non-legal" issues stemming from the complex, interactive environment of everyday life (Salmi-Tolonen, 2008, p. 1). This requires obtaining a holistic, bird's-eye view to legal phenomena and the social environment from which they emerge.

The multi-perspective orientation implies that proactive law is essentially a systemic approach. Despite the acknowledged connection between proactive law and systems thinking, however, the potential to develop proactive law theory by leveraging insights from systemic approaches has not gained much attention in previous proactive law research. Research in systems thinking and law has in general been comparatively limited (Pierson-Brown, 2020, p. 523). More understanding would be needed, especially on how to turn different theories into improved legal practices (Berger-Walliser, 2012, p. 311). The aim of this article is to contribute to filling this research gap by introducing the concept of systems intelligence in the context of proactive law, and by exploring what role systems intelligence might play in building proactive law competence. This exercise is conducted in the following three sections drawing on interdisciplinary literature, including socio-legal theory and systems theories. The following section will provide a brief overview of the origins of systems thinking and how it has influenced the contemporary conceptualizations of law, giving rise to many systems-aware legal approaches, such as proactive law. Section 3 introduces the concept

of systems intelligence and discusses its promises and tensions within the proactive law framework. This section illustrates how formal, problem-based and legal-system-centric practices of law tend to create holding-back systems called "reactive-law-generators", which potentially reproduce and uphold the systemic challenges society strives to overcome. Next, Section 4 illustrates how mitigating the negative impacts of the "reactive-law-generators" should be the primary focus of proactive law and proposes practical approaches to address this challenge by leveraging systemic insights through design methods. The paper will conclude with a summary of the key insights and some suggestions for further research on the topic.

2. Understanding law as a system

Applying a systems lens to observe legal phenomena is an old idea. However, it appears to have gained increasing interest over the past few decades, despite falling "out of vogue" in some academic circles during the 1990s (LoPucki, 1997, p. 483). It is possible to name at least two reasons for this. The first is the global polycrisis. The growing awareness of how the various types of sustainability challenges are driven by similar mechanisms has not only established systems thinking as a unified, interdisciplinary framework to understand these challenges, but has also created a demand for more nuanced and informed systemic approaches to law (Pierson-Brown, 2020, p. 523). This may explain the emergence of such approaches as complexity theory and law (Murray, Webb and Wheatley, 2019, p. 4), reflexive environmental law (Ross and de Almeida, 2024, p. 3), proactive law (Salmi-Tolonen, 2008, p. 1; Berger-Walliser, Shrivastava and Sulkowski, 2016, p. 10), people-centered justice (OECD, 2021, p. 20) and legal design (Doherty *et al.*, 2021, p. 3).

The second reason is technological development. Understanding complexity requires technology that enables collecting vast amounts of data, processing it and mapping it to human-comprehensible forms. Various novel technologies are helping lawyers to digest an increasing volume of information and the complexity of legal systems in ways that were not possible a decade ago. Databases, software programs, large language models and other digital aids are thus not only facilitating, but also driving systems research. It has been argued that without such tools, conducting robust empirical research on complexities of law would be impossible (Ruhl and Katz, 2019, p. 23).

Adopting a profound systems perspective to law seems to also reflect a more general philosophical turn. Capra and Luisi (2014, p. 12) describe how the early twenty-first century is being shaped by an intellectual change in paradigms and metaphors, moving from a mechanistic and atomistic understanding of the world to a worldview of self-organizing systems. Since the first general systems theory was introduced in the 1940s by biologist von Bertalanffy, various other fields of science have incorporated the systems analogy to develop their own theoretical and methodological frameworks, including computer science, engineering, automation, design, healthcare, business and management (von Bertalanffy, 1968; Buchanan, 2019, p. 86; Saffo, 1999, p. xvii; Meadows, 2008, p. 1). Johnston suggests that this systemic shift would be part of humanity's cultural maturity, the "new common sense" (Johnston, 2021, p. 164).

Von Bertalanffy (1968, p. 9) postulated that system-ness is not just a characteristic of biological organisms but of the organization of the universe, including artificial and social systems. Sharing the same vision, Norbert Wiener (195, p. 15) developed a field of systems thinking in the 1950s called *cybernetics* to address problems of communication and control in the human, animal and the machine. Later, in the 1970s, biologists Maturana and Varela (1998, p. 48) introduced *autopoietic* organisms, that is, systems that make themselves through self-referential feedback loops. The autopoiesis theory enhanced the general understanding of systems by observing that it is the participants within a system that create the very system itself, making the autopoiesis theory a useful analogy also for explaining the system-ness of social phenomena (Mobus and Kalton, 2015, p. 40), including law.

Drawing on the concept of autopoiesis, the German sociologist Luhmann (2004) built a comprehensive theoretical framework to observe law as a self-organizing, communication-based social system. According to Luhmann, society consists of various subsystems, such as law, economy, politics, media and science. For each subsystem, other systems form its environment. Each subsystem creates and recreates itself by distinguishing and comparing information from their environment to their internal information using communicational binary codes that are specific for each system. For example, the legal system identifies itself by coding what is "legal" or "illegal", and the economic system by what is worthy of "payment" or "no payment" (Luhmann, 1989, p. 45). Each system also uses their own mediums, programs and structures to process information from their environment. This means that the legal system is operationally closed. Law has its own (legal) institutions, particularly courts, that hold the authority to communicate the meaning of

law. In Luhmann's (2004, p. 11) view, the purpose of law's communicative processes is to produce normative expectations for society, allowing people to anticipate future events and prepare for disappointments.

Developing Luhmann's idea of law as an autopoietic system, Teubner (1983, p. 257) has argued that modern law is inherently reflexive, as the function of the legal system is to "regulate the self-regulation" of other systems. This refers, for instance, to regulation through businesses' self-regulatory practices and disclosure- and procedure-based instruments, which are also recognized as potential proactive law steering mechanisms in proactive law literature (Berger-Walliser, Shrivastava and Sulkowski, 2016; Pohjonen, 2006, p. 59). Furthermore, building on the systems theoretical frameworks of both Luhmann and Teubner, Kjaer (2022) has proposed a concept of transformative law. Kjaer suggests that contemporary law should be understood particularly through its form-giving function. Therefore, the legal system would not only uphold normative expectations, but also transform social phenomena through its communicative processes (Kjaer, 2022, p. 779). The idea of transformation through law is also familiar to proactive law, which emphasizes scanning for opportunities for positive change and setting change-oriented goals in legal work (Berger-Walliser, 2012, p. 29).

Luhmann's theory has been criticized for being too abstract and for decentring humans as conscious agents of social systems. Also, according to critics, the use of autopoiesis as a key conceptual element to describe the systemic nature of law is merely a "theoretical artefact, not of social observation" (Murray, Webb and Wheatley, 2019, p. 6; LoPucki, 1997, p. 483). It is also questionable to what extent can theories developed in natural sciences be used in describing social phenomena (Teubner, 1992, p. 25). However, the benefit of Luhmann's systems theory is that it makes it psychologically easier to comprehend and discuss the abstract and complex nature of law. It can also help navigate the often dualistic conceptualizations of law in modern jurisprudence, where law is either observed as a closed (•), purely normative system of rules and legal procedures, or as an open (≈) system that recognizes the organic nature of law, its ambiguity and complex relations to society. This typology is illustrated in Table 1. In the light of systems theory, it becomes possible to observe law as a system that can be both, or more precisely, as a system that depends on both closedness and openness in its operations. As noted by Aarnio (1987, p. 5), there seems to be a natural tension between these two aspects of law, which also explains the challenges of legal decision-making and problem solving.

Table 1: Law as a closed/open system

	Law as a closed (•) system	**Law as an open (≈) system**
Perspective to the legal system	Internal view of the legal system, one "legal" world paradigm, law as an "abstraction"	External view of the legal system, legal realism, legal pluralism, many legal worlds, law as a "practical reality"
Approach to normativity	Formal state law, hard law, legal norms, norms of legal practices and procedures	Social norms and behavioral norms influencing legal norms, informal law, private and customary legal practices, soft law
Legal research traditions	Classical legal dogmatics, legal positivism	Critical legal studies, socio-legal and interdisciplinary legal research traditions
Emphasis in the rule of law	Formalism, legalism, predictability, legal certainty	Functionality, reasonability, purpose, (social) justice

In Luhmann's theory, the closedness of the legal system refers to the autopoietic, self-referential processes that happen within the legal system. It is these processes that draw the borders of law. In jurisprudence more generally, the closedness of the legal system refers to the desire to define clearly what law is. This approach is emphasized particularly in Kelsen's (1989) "pure theory of law", but also by other classical legal positivists like Hart (1994) and Raz (1979, p. 38). The closedness serves particularly practical legal interpretation. A judge deciding a case needs to be able to operate within a system that is recognizable and definable, to avoid arbitrariness and misuse of power. Therefore, for logical reasons, law as a closed system focuses primarily, though not exclusively, to the internal perspective of the law, and emphasizes legalism and production of normative expectations as the moral rational basis for the rule of law and legal certainty (Aarnio, 1987, p. 5). This internal perspective includes legal practices, particularly court proceedings and other judicial processes (Tuori, 2002, p. xi), as well as physical "law-related objects" like courthouses, judges, lawyers, law books and legal documents (LoPucki, 1997, p. 488).

In contrast, when law is observed as an open system, there is no similar intention or need to define clearly what law is (or "ought to be"), but rather a need to understand law more holistically as a multidimensional social phenomenon. In Luhmann's theory, the openness of the legal system means that law is *cognitively open*. Cognitively open systems can learn from their

environment by reflecting the information they receive from other systems. Whereas closedness of the legal system is important for the identification and definition of law, cognitive openness serves the need of information exchange and integration of new (legal) knowledge into the legal system for the development of law (Luhmann, 2004, p. 79).

In legal research, the openness of law enables obtaining "outer", pluralistic and critical perspectives to law that seek to reveal and address the social, economic, cultural and behavioral norms behind the legal norms. This orientation is typical to now-traditional legal research fields such as critical legal studies, legal realism, feminist jurisprudence and socio-legal jurisprudence (Murray, Webb and Wheatley, 2019, p. 11; Nedelsky, 2011, p. 19; Pohjonen, 2006, p. 58) as well as more interdisciplinary approaches such as behavioral law and economics, anthropology of law and psychology of law (Pirie, 2013, p. 4; Zamir and Teichman, 2018, p. 145). At a practical level, openness can refer to recognizing the role of many informal and customary practices in serving real-life legal and justice needs (OECD, 2021, p. 17), or paying attention to pragmatic concerns in business life, such as how the availability of financial resources and business relationships shape companies' approach to legal matters (Haapio, 2006, p. 26). Complementing legal formality and predictability as the central elements of rule of law, the open system perspective emphasizes justice, reasonability and proportionality in legal solutions (Aarnio, 1987, p. 7).

Proactive law literature highlights the need to approach law as an open system, yet with respecting the internal coherence of law, to ensure legal certainty and the fulfilment of rule of law (Pohjonen, 2006, p. 57; Berger-Walliser, 2012, p. 28). Pohjonen and Koskelainen (2012, p. 270) have used the metaphor of the blind-folded goddess of law to describe how law, instead of operating like a *neutrally blind goddess*, should rather become a *seeing goddess*, who "is aware of her subjectivity" and "takes responsibility of her influence on creation of laws and their implementation", and who can "see the variety in the world and know that it cannot be managed with monolithic methods". Systems theories offer a way to make this vision and attitude practicable. Systems theory, such as the one by Luhmann, can work as a mental framework or a thinking tool that allows "viewing" in and out of the legal system while maintaining the boundaries of different systems and acknowledging the possible tensions between opposite perspectives (Hietanen-Kunwald, Haapio and Toivonen, 2021, p. 72). Systems thinking can thus help "the goddess of law" see causalities and potential solutions beyond the limits of legal formalism and develop legal solutions that better

address the needs of the people. It must be noted, however, that such reflexive observations at the systems level may not be enough to actualize the desired solution or generate more systemic change in practice. As highlighted by Pohjonen (2006, p. 63), ultimately self-reflexion must be practiced in real life by those who participate in creating the social systems of law. This is a task where the systems intelligence framework can become useful.

3. Understanding oneself as part of the system

From a practical perspective, the limitation of systems thinking and systems theories is that they do not offer much guidance on how to apply systemic insights in practice. For instance, Luhmann's systems theory focuses on describing law as an abstract communication-based system, but particularly excludes humans as active agents who produce the communication. The abstract and difficult formulation of the theory itself may also explain why it has not inspired much empirical research (Pierson-Brown, 2020, p. 523). In general, the externalist and objectifying orientation of systems thinking is said to highlight abstract, cognitive rationality instead of rational action and pragmatism (Hämäläinen and Saarinen, 2006, p. 191). This is also something that seems to characterize traditional legal scholarship (LoPucki, 1997, p. 489). Therefore, to develop proactive law competence with systems insights, it is useful to explore more hands-on approaches that help bridge the gap between systems theories and practice.

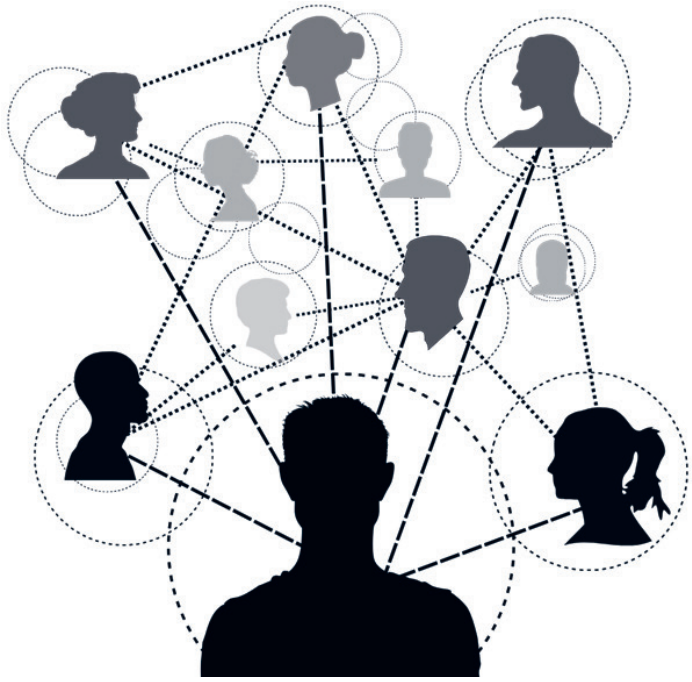
One possible approach is systems intelligence, a practical philosophy of acting intelligently within complex systems, initially introduced by Raimo Hämäläinen and Esa Saarinen (2004) at Aalto University in Finland. Systems intelligence is based on the framework and key concepts of systems thinking, but seeks to reach beyond it in its proactive, pragmatic, personal and existential emphasis (Hämäläinen and Saarinen, 2004, p. 51). Similarly to proactive law, systems intelligence is inspired by various fields of holistic thinking and human sciences, such as theories on human decision-making and problem solving through positive psychology. Both approaches emphasize that when resources are allocated to efforts that support a goal, rather than to efforts that seek to avoid an undesirable outcome, the goal is more likely reached (Hämäläinen, Jones and Saarinen, 2014, p. 199; Berger-Walliser, 2012, p. 27). Systems intelligence also draws on situation analysis and philosophical practices, such as the Socratic tradition in philosophy, which

emphasizes conceptual thinking for the creation of good life (Hämäläinen and Saarinen, 2004, p. 52). Furthermore, systems intelligence builds on the five disciplines of a learning organization developed by Senge, particularly on Senge's (2006) views of personal mastery and systems thinking in organizations.

Unlike systems theories, systems intelligence does not observe systems as mere conceptual abstractions that exist outside of their human observers, but places humans at the centre of the systems and considers them as their integral parts. Systems intelligence also fosters a flexible understanding of what a human system is. For Hämäläinen and Saarinen (2004, p. 53), a human system can be a company or global economy, as well as a business relationship, a meeting or a random gathering with friends. In systems intelligence, a human system is therefore *any generative frame within which a subject can experience their reality as taking place.* A person acts in a systems intelligent way when they can engage with this frame successfully, contributing to sustainable outcomes that ideally benefit everyone within the frame (Hämäläinen and Saarinen, 2006, p. 191).

Both systems intelligence and the proactive law theory highlight that to generate positive outcomes, people must first become aware of the systems they are part of and position themselves within them (Hämäläinen and Saarinen, 2006, p. 191; Pohjonen, 2006, p. 63). For example, an in-house lawyer can typically observe themselves as part of a legal team or a managerial unit, as well as part of the whole organization where they work. They can sense the invisible social norms that form the organization's culture and may intuitively align their own behaviors to match that culture or try to intentionally act against it. A systems aware in-house lawyer can also perceive other systems that interconnect and overlap with their own systems. In other words, a systems intelligent lawyer is aware about other perspectives beyond the boundaries of the legal system and is willing to tune into them through a wide range of communications, not just those of law (Pohjonen, 2006, p. 55). This involves using not only legislation and other authoritative legal sources to address legal issues but seeking useful information and solutions also from other professional fields. Therefore, a systems intelligent lawyer understands the impact of making connections and working together with other types of experts who contribute to the system's wellbeing, including clients, business partners and government agencies (Haapio, 2006, p. 26). Similarly to systems intelligence, Edson, Tuddenham and Buckle-Henning (2016, p. 16) call this kind of systemic awareness *systemic sensibility*, which they describe as "an innate, intuitive or tacit appreciation of systemicity in the empirical world".

Figure 1: A systems intelligent lawyer can perceive themselves as part of interconnected human systems and act purposefully within them



However, according to systems intelligence, mere systems sensitivity is not enough to be regarded as systems intelligent behavior nor to generate positive systemic impacts. In addition to being aware of the systems around them, people must also be able to perceive themselves as capable of contributing to them and actively choose to do so (Hämäläinen, Jones and Saarinen, 2014, p. 20). For example, a lawyer who behaves systems intelligently can maintain positively engaging feedback relations within the systems they are part of, scan opportunities for promoting desired goals, act timely and effectively to prevent risks and damages and create new solutions to address legal problems (Berger-Walliser, 2012, pp. 29–30).

Hämäläinen and others point out eight factors of systems intelligent behaviors that people can continuously develop. The authors argue that these factors can also be used to assess one's personal systems intelligence. A systems intelligent test based on these factors, provided by the Systems Intelligence Research Group (no date) has provided a systems intelligent test based on these factors (Hämäläinen, Saarinen and Törmänen, 2019, p. 643). The eight systems intelligence factors are:

1) *Systemic perception*: Ability to see, identify and recognize systems, patterns and interconnections around oneself, and to have situational awareness;
2) *Attunement:* Ability to engage intersubjectively, be present, mindful, situationally sensitive and open;
3) *Positive attitude*: Ability to keep a positive outlook, to not get stuck on negative impressions and effects;
4) *Spirited discovery*: Ability to engage with new ideas and embrace change;
5) *Reflection*: Ability to reflect upon one's thinking and actions and to challenge one's own behavior;
6) *Wise action*: Ability to exercise long-term thinking and realize its implications, understanding that consequences may take time to develop;
7) *Positive engagement*: Ability to put systemic leverage points and means successfully in action with people;
8) *Effective responsiveness*: Ability to put systemic leverage points and means successfully into action with the environment, to "dance with systems".

Mastering such systems intelligence abilities is also recognized in proactive law literature as a central proactive law competence, albeit acknowledging the challenges they pose. The paradox of proactive lawyering is that, to act proactively, lawyers must first have enough knowledge about the non-legal context and, vice versa, their clients must possess enough knowledge about the law (Haapio, 2006, p. 31; Berger-Walliser, 2012, p. 29). Furthermore, each stakeholder needs to be aware of their personal feelings, attitudes and motivations which influence their ability to initiate interaction and respond to others' initiatives—as well as empathize with those of others (Pohjonen, 2006, p. 63). The decisive moment for any proactive legal action is when to reach out to ask for, or offer, legal guidance. From the viewpoint of systems intelligence, this requires being open and attuned to the various signals from the environment that indicate when and what action is needed to improve a particular situation and respond to them with good intentions (Hämäläinen, Jones and Saarinen, 2014, p. 21). Reaching such a level of informed and emotionally intelligent working culture in which proactive collaboration is possible takes time and personal effort from all stakeholders.

Systems intelligence does acknowledge that unleashing the potential of more advanced systems intelligence competences requires continuous

learning, particularly through practice (Hämäläinen, Jones and Saarinen, 2014, p. 15). Similarly, Alhanen (2016, pp. 206–206) has argued that people can develop their "sense of inclusion" (in Finnish *osallisuustaju*) only over time through constructive dialogue with people who represent different perspectives, rather than master such competences naturally. Edson, Tuddenham and Buckle-Henning (2016) propose that the more advanced systems abilities beyond systemic sensitivity would be *systems literacy* and *systemic capability*. In contrast to Hämäläinen and others, they argue that these skills would be acquired through education in systems sciences and systems thinking, which would give a person "clear concepts and a common language" to articulate and reflect on their innate systemic sensibility "and act on it in a considered way" (Edson, Tuddenham and Buckle-Henning, 2016, p. 16).

Drawing on Senge's (2006, p. 136) first discipline, *personal mastery*, the systems intelligence philosophy emphasises that becoming more systems intelligent is an internal process of an individual that happens through the mechanism of self-reflection. Hämäläinen, Jones and Saarinen (2014, p. 22) propose that reflecting one's own behavior and beliefs can help override the automatic cognitive processes that typically limit seeing the bigger picture, adopting new perspectives and tuning into possibilities for change. This reminds the logic of autopoietic, cognitively open systems adopted in Luhmann's systems theory, which is further developed by Teubner (1983) in his reflexive law approach. According to systems theory, social systems can only change through their internal processes, which are self-reflective (Teubner, 1983, p. 272). Proactive law literature also recognizes the importance of developing "good levels of self-knowledge" to establish a "dynamic balance" in collaborative situations and to consider how oneself can "create the preconditions for achieving goals in a manner that does not contribute to unnecessary problems" (Pohjonen, 2006, pp. 55, 63). Similarly, Senge (2006, p. 21) points out that "true proactiveness comes from seeing how we contribute to our own problems". Good self-reflection skills thus may not only help act more systems sensitively in different contexts, but also help avoid systemic traps.

4. Breaking the reactive-law generators with systems intelligent proactive law

Although human systems generate themselves naturally, and most emerge for a well-meaning purpose, such as a business relationship or a lunch group, systems intelligence notes that people can also get caught in systems that serve nobody's interest (Hämäläinen and Saarinen, 2004, p. 52). Typically, these paradoxical systems are rooted in misunderstandings, disinformed beliefs and erroneous assumptions about how one should behave or what kind of behavior is anticipated from other people. Our inner knowledge and intuitions, the noncognitive elements that are crucial also to systems intelligent behavior, can also lead us astray (Pronin, 2008). Hämäläinen, Jones and Saarinen (2014, p. 31) call such phenomena "systems of holding back", which refers to people's systemic bias to not act positively and constructively when engaging with others. The authors suggest that even if people can "think in systems" and "see the big picture", they will not be able to generate any positive systemic change if they do not simultaneously pay attention to the "hidden, crushingly powerful counterforce" of holding back systems (Hämäläinen and Saarinen, 2006, p. 199). Meadows (2008, p. 111) refers to such systemic forces as *systemic traps*. Similarly, Johnston (2021, p. 331) calls them *polar fallacies*, emphasizing how systemic challenges reflect the underlying cognitive and psychological patterns of binary thinking.

In systems of holding back, people often display behaviors they would change if they only could have a holistic view of the setting they are in, involving the perspectives of others about the given situation. According to Hämäläinen and Saarinen (2004, p. 54), this is *the paradox of human systems*. One example of such negative behavioral loops mentioned by the authors is the waste-generating system that our industrialized world desperately tries to overcome. While consumers, businesses and governments would like to produce less waste, more and more waste is produced. Another example of human paradoxes are systemic non-support generators taking place in many workplaces. While managers would like to give more support for their employees, who also wish greater support from their managers, the outcome is not increased support (Hämäläinen and Saarinen, 2004, p. 54). Similarly, MacLoud (2024) describes how human systems can create both beneficial, but also unintended and unexpected "emergent properties". For instance, while the purpose of the justice system is to protect people from criminal activities, it can also establish systemic mechanisms that marginalize people (MacLoud, 2024, p. 118).

It is easy to view the reactive way of operating law as another example of paradoxical human systems. Many possible examples could be given to illustrate the point. In the context of an organization and its in-house legal team, a relevant example could be the absence of sufficient and timely legal support for in-house clients. Lawyers wish that their in-house clients would ask legal help and try to solve their legal needs before any problems arise, not just to make the organization function better, but also to make the legal work more meaningful and manageable. At the same time, the in-house clients wish they could get their work right by getting legal guidance before any mistakes occur. Yet the very opposite tends to happen. Instead of acquiring legal help to address avoidable legal risks in advance, people react once a problem arises, or, in the most extreme case, contact a lawyer when a "fighter" is needed for a courtroom battle (Haapio, 2006, p. 29). Berger-Walliser (2012, p. 24) notes that the human tendency to react to problems only once they have occurred has created a legal culture where reactive problem solving is normalized and preventive or proactive problem solving is perceived as its "alternative" or abnormal. Drawing on the insights of systems intelligence, this paradox of human systems could be called *the systemic reactive-law generator*.

However, from a systems perspective, it becomes evident that it can be particularly challenging for both lawyers and their clients to resist the systemic structure and real-life settings of "the reactive-law generators". Systems theories describe how it is the systems that eventually cause their own crises through repeating similar behavioral patterns, not external forces nor single mistakes (Meadows, 2008, p. 2). Therefore, different people who are stuck in the same system are often compelled to produce qualitatively similar results (Senge, 2006, p. 40). In other words, the emergent property of "reactiveness" is not caused by individual players but depends on the way the system operates (MacLoud, 2024, p. 118).

In the case of the reactive-law generators, a contributing factor is the self-referential communication processes of the legal system that require an issue to be first recognized as "legal" before the legal system can properly deal with it (Toivonen and de Francisco Vela, 2024, p. 265). As described by Luhmann (2004, p. 17), the legal system is cognitively open to information from its environment, but any information that does not resonate with the inner logic or coding of law, such as issues of business relationships, financial interests or mental health, will be rejected by the system. Consequently, a lawyer will not be solving a case if they think there is nothing "legal" in it in a formal, closed-legal-system sense. From the perspective of formal

law, lawyers' particular role is to interpret the law, and thus participate in the communicative processes of the legal system in producing normative expectations (Luhmann, 2004, p. 17).

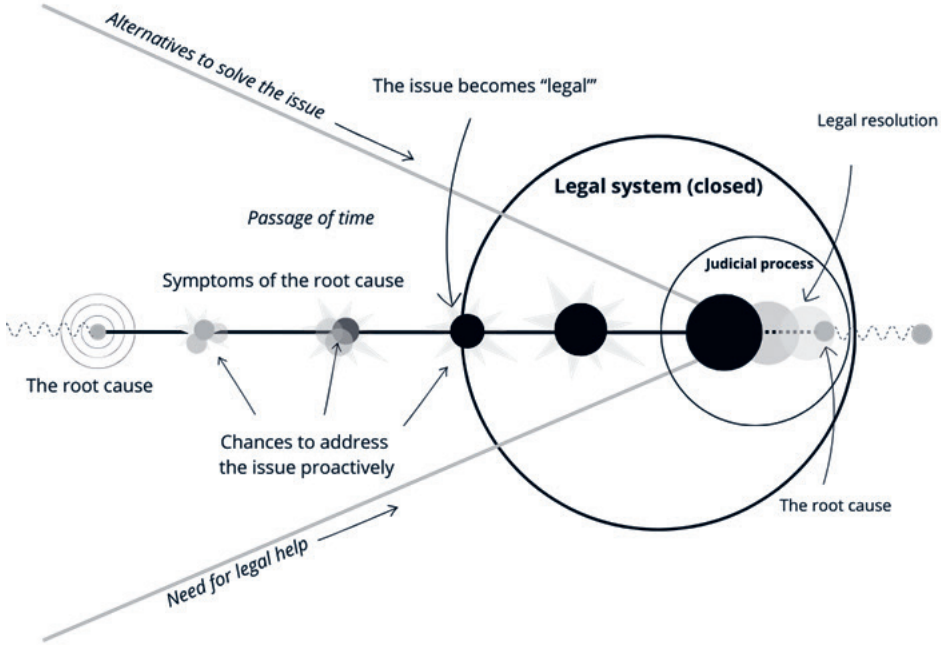
This mechanism can also have reverse effects outside the legal system, causing people to not contact lawyers early enough or practice proactive legal planning. When conducting their everyday tasks, people usually do not consider the legal aspects of their activities unless a specific "legal" reason arises. LoPucki (1997, p. 489) argues that in peoples' lives "law is marginal". This is especially the case when people are not familiar with the content of the law in a given context, and therefore have no necessary cognitive tools to imagine what the legal implications of their activities would even be. Pirie (2013) notes that people identify and conceptualize law differently, from the perspective of their previous personal experiences. Therefore, what people consider as "law" in the first place can differ significantly from the educated, formalistic understandings of lawyers. From an anthropological perspective, there is no one unified understanding of "legal" (Pirie, 2013, p. 8). Typically, the need for legal information or help stems from a negative event, such as from a conflict of interests, misbehavior or loss. When "legal" is associated with "problem", it fosters a shared culture where law is identified and addressed primarily through the negative lens of problems rather than the positive lens of opportunities.

To illustrate how such reactive-law cycles can develop in practice, we can again think of an in-house legal team and its in-house clients. From a systemic perspective, legal problems do not appear "out of nowhere", but stem from conditions that usually have been present for a longer period without anyone noticing or understanding their potential to develop into legal problems. Such a root cause can, for example, be a poorly arranged contract management system, lack of transparency in organization's decision-making, or communicating an organization's environmental protection policies to employees in unintelligent jargon. How such root causes can develop into legal problems is illustrated in Figure 2: If the organization is stuck in a system of holding back, the root cause remains ignored until obvious symptoms start to emerge. Each participant's personal systems blindness—failure to recognize their individual role in addressing the issue—amplifies the negative causality. Only when a crucial contract is lost, a decision appears biased, or an environmental policy is misunderstood or ignored, may people begin to consider the legal consequences of such events. However, the organization may still hold back from addressing the initial cause and taking action to tackle the holding-back system. Over time,

the issue has potential to develop into a more complex matter with fewer alternative, "non-legal" solutions available.

The tipping point on a legal problem escalation axis occurs when an issue becomes so evidently problematic that it is finally recognized as a "legal problem". For instance, when the organization neglects a clause of a missing contract, a director is suspected of exceeding the limits of their decision-making power, or an employee causes an environmental damage, there are legal rules and procedures telling how the issue should be handled. From a systems theoretical perspective, this means that the issue can now be solved within the operationally closed legal system, using "legal" means such as legal interpretation and judicial procedures. The in-house clients contact the in-house legal team, and resolving the issue becomes "the responsibility of the lawyers". The matter may even escalate into a full legal case, subject to court proceedings or other legal processes, depending on the nature of the issue. Eventually, the court or another judicial authority reviews the facts and applies the law to reach a legally rational decision.

Figure 2: Illustration of a legal problem escalation axis



However, the court may not be able to address the root cause of the problem if it falls outside the scope of "legal" matters (Hietanen-Kunwald, Haapio and Toivonen, 2021, p. 79). This means that the root cause may persist even when a "legal" resolution is achieved. If the organization fails to investigate the underlying cause, being stuck in the system of holding back, there is a risk that unresolved issues may give rise to new problems, perpetuating not only the reactive-law cycle but also other negative systemic paradoxes. Especially in the case of environmental damages, pollution and habitat destruction, the remedies provided by the reactive legal problem solving—monetary compensation or punishments—can never fully restore what is lost in biodiversity and ecological balance (Berger-Walliser, Shrivastava and Sulkowski, 2016, p. 10).

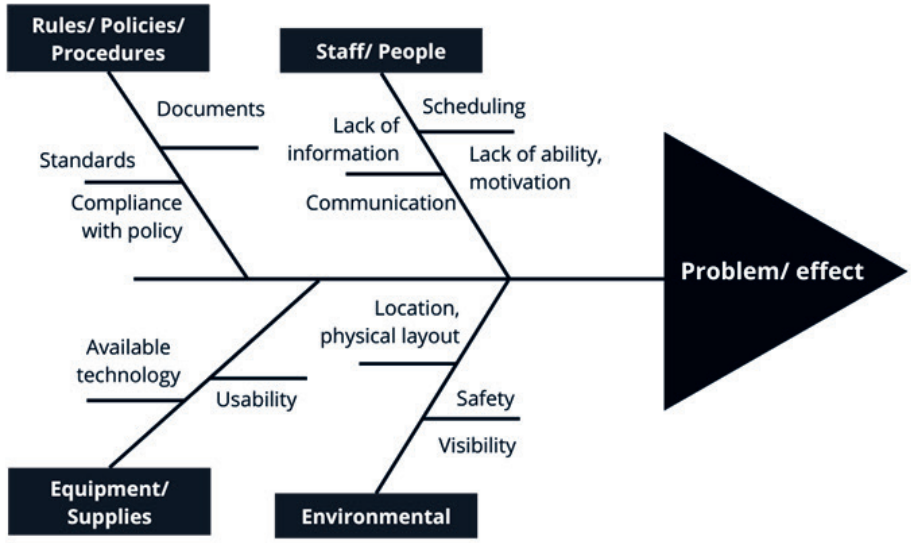
While court proceedings and other formal legal processes may have negative side effects and be unsatisfactory from a systemic perspective, they nevertheless offer a silver lining. According to Meadows (2008, p. 112), a system trap is always also a system opportunity. A court proceeding provides an opportunity for the organization to learn from the experience, turning the negative events and consequences into insights that can help break negative behavioral cycles. Pohjonen and Koskelainen (2012, p. 273) call this kind of learning opportunity *legal autopsy*. Careful retrospective analysis of the root causes of the issue that ended up in court, conducted ideally through a participatory and dialogical process involving all key stakeholders, has the potential to reveal deficiencies, misunderstandings and hidden motivations within the organization's operations and create collective understanding of what needs to change. This reactive, yet transformation-seeking orientation aligns with the idea of proactive law, as well as with the concept of systems intelligence, both of which emphasize self-reflection and collective learning, collaboration, effective responsiveness and positive interaction in the face of setbacks (Pohjonen, 2006, p. 62; Hämäläinen, Jones and Saarinen, 2014, p. 15).

Breaking the above-described cycles of the "systemic reactive-law generators" can be considered one of the main goals of proactive law (Berger-Walliser, 2012, p. 27). To reach this goal, it is necessary to understand both why people end up in systems that make them present undesirable behaviors, but also how to change a system that does not satisfy the needs of the people who are living in these. Systems intelligence philosophy suggests that these epistemic objectives could be reached simultaneously through the development of individual and collective systems intelligence competence (Hämäläinen and Saarinen, 2004, p. 54).

Although Hämäläinen and Saarinen (2004) underscore that systems intelligence is not about systems mapping or illustrating cycles and loops, but rather an ability that an individual can master in everyday social interactions, many visual, human-centric and collaborative methods can potentially help develop systems awareness, ultimately leading to improved systems intelligent behavior. Hämäläinen and Saarinen (2004, p. 52) point out that systemic awareness can particularly be created by expanding one's own individualistic perspective by observing the situation "through the eyes of the other", which can also help grasp features of the system, and understanding how everyone inside the system contribute to creating it. This is particularly important in cases of dispute management and prevention, as people tend to prioritize their own objectives for a conflict settlement (Hämäläinen, Jones and Saarinen, 2014, p. 113). Similarly, MacLoud (2024, p. 119) notes that when a lawyer needs to act, the holistic perspective provides insight into why legal challenges or opportunities emerge the way they do, and whose experiences need to be taken into consideration when creating solutions. Further, Johnston argues that overcoming dualistic fallacies, like "us" versus "them", requires more hands-on approaches that have integrated the intellectual meta-perspective of systems thinking (Johnston, 2021, p. 360). Various design methods are recognized as efficient aids in this work (Hämäläinen *et al.,* 2020; Buchanan, 2019).

For instance, visual system analysis techniques applied as part of multi-stakeholder design workshops can make each and every person's role and perspective in the system more visible—and therefore more understandable to others (Berger-Walliser, Barton and Haapio, 2017, p. 31). Visualizing patterns and connections also helps identify systems and show their complexity. Meadows (2008, p. 5) notes that because "systems happen all at once", it is necessary to use pictures to describe all their elements and parts in an adequately understandable way. Also, when the systemic nature of an issue becomes visualized, it can help spot the critical factors in interactions and relationships that contribute to problematic systemic behaviors. One possible technique is the classical fishbone model by Kaoru Ishikawa, shown in Figure 3: The fishbone diagram can be used, for example, to build collective understanding within an organization about how various seemingly innocent and unrelated events can escalate into legal problems, which helps prevent them in the future. The problem or effect is displayed at the "head" of the fish and contributing events and their root causes are listed on the "bones" under different cause categories.

Figure 3: Example of a "fishbone" root cause analysis diagram (Source: Author's original illustration inspired by Ishikawa and Loftus, 1991)



5. Conclusion

We live in a world of systems. Some systems emerge organically, while others are made by us, humans. In modern jurisprudence, law is observed as a system, though its nature has been widely debated. Legal problems are also characterized as being systemic, arising from complex, deep-rooted social structures, human interactions and relationships. The challenge with solving systemic problems is that they will persist until their root causes are adequately understood and addressed. From the viewpoint of proactive law, the only sustainable way of solving systemic problems is to approach them proactively rather than reactively. Proactive law focuses on preventing future problems and promoting desirable outcomes by using creative, collaborative and alternative methods to traditional legal problem solving, while striving for legal certainty and rule of law. This article explored how systems intelligence, a concept and pragmatic philosophy developed by Raimo Hämäläinen and Esa Saarinen, could help develop proactive law competence, and thus potentially support proactive legal problem solution and problem prevention.

As described in this paper, systems intelligence and proactive law share many similar attitudes, principles and philosophical perspectives. Both approaches emphasize systems-sensitive capabilities like self-reflection, collaboration skills, and a sense of individual responsibility to contribute to the whole, in other words, the "consideration of how oneself can create the preconditions for achieving goals in a manner that does not contribute to unnecessary problems" (Pohjonen, 2006, p. 55). However, it has not been discussed in previous proactive law literature how people can develop such skills. If an individual does not perceive themselves as having the power or ability to make a difference, they can hardly act in a proactive way. Therefore, a key insight from the systems intelligence framework for proactive law is that proactive lawyering can be viewed as a competence similar to systems intelligence. This competence can be actively developed through practical systemic methods, such as systems intelligence itself or visual and collaborative design approaches.

Furthermore, systems intelligence provides a relatable, human-centric perspective for observing the legal system and its feedback mechanisms. Unlike abstract systems theories, systems intelligence places humans within systems and seeks to understand how different behaviors influence them. According to systems intelligence, human interaction tends to create systems of holding back, which can develop over time into complex systemic paradoxes—the ongoing sustainability crisis being one example. Building on this idea, this article illustrated how the formalistic and problem-focused practices of law, representing the idea of law as a closed system, can produce systemic "reactive-law generators". Developing a more nuanced understanding of these reactive-law generators, their mechanisms, and strategies to mitigate their negative effects is a task for future proactive law research.

Applying systems intelligence to develop proactive law competence is, of course, still a potentiality rather than an empirically tested practice. However, empirical studies on systems intelligence offer insights into the promises of systems intelligence in practice. Hämäläinen, Saarinen and Törmänen (2019, p. 643) report that systems intelligence, as assessed by peers using the eight systems intelligence factors introduced in this article, correlated with higher job performance among a cohort of 2,060 individuals, including university students, engineering company employees, daycare workers and managers. The finding is supported by another study which identified a positive relationship between organizational systems intelligence and organizational wellbeing and performance (Jumisko-Pyykkö *et al*., 2022, p. 60). Additionally,

a case study demonstrated that a systems intelligence card game supported systems intelligence learning among teams of early childhood professionals (Hämäläinen *et al*., 2020, p. 166). Similar studies conducted on lawyers would present an interesting research opportunity for developing proactive law.

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